

Midterm Lab Task 3 - Python List Collections

Problem 1. Using List Collection type. Create a program that will allow the user to perform the following **functions**: (add, update, search, delete, display, and sort) items in a list:

Note: You are free to decide what data you will be storing in the list and name the list based on the type of data you wish to store.

[MENU OPTIONS]

- 1 – Add Items**
- 2 – Search for an Item**
- 3 – Remove an Item**
- 4 – View all items (Sorted either A-Z | Z -A)**
- 0 – Exit program**

Pick one [0 to quit]: ____

Requirements:

1. The user can add items in the list until the user presses x to stop
2. The user should be able to perform **search** if an item exists – Display if found or not found and count the number of instance in the list.
3. The user should also be given the option to remove an item in the list – Display the Message “Item found and deleted” once deletion is performed – else display “item not found-deletion unsuccessful”
4. The user may also opt to view items in the list and display items sorted in Ascending order
5. The user may opt to exit the program by typing 0

Note: you are free to design the interface of the program, base on the Menu options shown.

Code:

```
items = []

def menu():
    while True:
        print("\nWelcome to 700P ProgramCart")
        print("1 - Add Items")
        print("2 - Search for an Item")
        print("3 - Remove an Item")
        print("4 - View all items")
        print("0 - Exit program")

        choice = input("Enter choice [0 to quit]: ")

        if choice == "1":
            while True:
                newItem = input("Enter item to add (or 'x' to stop): ").upper()
                if newItem == "X":
                    break
                items.append(newItem)
                print("Items added successfully.")

        elif choice == "2":
            search = input("Enter item to search: ").upper()
            count = items.count(search)
            if count > 0:
                print("-----")
                print(f"Item '{search}' found {count} time(s) in the list.")
                print("-----")
            else:
                print(f"Item '{search}' not found.")

        elif choice == "3":
            remove = input("Enter item to remove: ").upper()
            if remove in items:
                items.remove(remove)
                print(f"Item '{remove}' found and deleted.")
            else:
                print(f"Item '{remove}' not found - deletion unsuccessful.")

        elif choice == "4":
            if len(items) == 0:
                print("The list is empty.")
            else:
                print("\n--- Items in the List ---")
                for i, item in enumerate(items, 1):
                    print(f"{i}. {item}")

        elif choice == "0":
            print("Exiting program... Goodbye!")
            break

        else:
            print("Invalid choice, try again.")

menu()
```